

scripts/check-canonical-root-and-upstreams.sh — User Guide

Last verified: 2026-05-15 (constitution-compliance plan Phase 8)

Inheritance: HelixConstitution §11.4.35 (Canonical-Root Inheritance Clarity) + §11.4.36 (Mandatory install_upstreams) + §11.4.18 (script docs)

Overview

Combined verifier for two related HelixConstitution clauses:

§11.4.35 — Canonical-Root Inheritance Clarity

Three sub-checks:

- **(a)** Consumer's root `CLAUDE.md` and `AGENTS.md` MUST open with an inheritance pointer (within the first 40 lines): either `## INHERITED FROM constitution/<file>` heading OR Claude Code's `@constitution/<file> import` syntax.
- **(b)** The constitution submodule's three canonical files MUST exist: `constitution/CLAUDE.md`, `constitution/AGENTS.md`, `constitution/Constitution.md`.
- **(c)** The constitution submodule's own `CLAUDE.md` and `AGENTS.md` MUST NOT carry `## INHERITED FROM` at top level — they ARE the canonical root and must not appear to inherit from elsewhere. **Pattern is allowed inside fenced ``` code blocks** (where it's documentation showing consumers the inheritance pattern).

§11.4.36 — Mandatory install_upstreams script

Every owned-by-us submodule MUST ship an `install_upstreams` script at one of these locations:

- `<submodule>/install_upstreams.sh`
- `<submodule>/install_upstreams`
- `<submodule>/scripts/install_upstreams.sh`
- `<submodule>/scripts/install_upstreams`
- `<submodule>/Upstreams/install.sh`
- `<submodule>/upstreams/install.sh`

The script is what a freshly-cloned submodule uses to self-provision its own remote topology (per §6.W: github + gitlab minimum for vasic-digital submodules; 4 mirrors for HelixDevelopment-owned submodules).

Per-submodule waivers are supported (INSTALL_UPSTREAMS_WAIVERS=() in the scanner) with mandatory rationale. No active waivers in 2026-05-15 baseline.

Modes

Flag	Behavior
--strict (default)	Exit 1 on any violation
--advisory	Exit 0 even on violation
LAVA_CANONICAL_ROOT_STRICT=0 env	Same as --advisory

Usage

```
bash scripts/check-canonical-root-and-upstreams.sh
# strict (default)
bash scripts/check-canonical-root-and-upstreams.sh --advisory
# exit 0 even on violation
LAVA_CANONICAL_ROOT_STRICT=0 bash scripts/check-canonical-root-
and-upstreams.sh # env-driven
```

Hermetic test

tests/check-constitution/test_canonical_root_and_upstreams.sh — 6 fixtures:

1. test_compliant_fixture_passes — baseline compliant fixture
2. test_missing_root_pointer_rejected — root CLAUDE.md without inheritance pointer
3. test_canonical_self_inheritance_rejected — canonical root carrying its own ## INHERITED FROM outside any code block
4. test_missing_install_upstreams_rejected — submodule lacking install_upstreams
5. test_advisory_mode_returns_zero — advisory swallows exit code
6. test_canonical_fenced_code_block_pointer_passes — ## INHERITED FROM INSIDE a fenced ```markdown block is documentation, not violation (discrimination test for false-positive prevention)

Known violations (2026-05-15 audit)

§11.4.35 sub-checks all PASS in current Lava tree (after the Phase 8 false-positive fix in §11.4.35.c — the constitution submodule's documentation of the pointer pattern inside a fenced code block is correctly recognized as non-violating).

§11.4.36 — 10 of 16 owned submodules MISSING `install_upstreams`:

Submodule	Status
Auth	✓ has <code>install_upstreams</code>
Cache	✓ has <code>install_upstreams</code>
Challenges	✗ MISSING — owed in follow-up
Concurrency	✓ has <code>install_upstreams</code>
Config	✗ MISSING — owed in follow-up
Containers	✗ MISSING — owed in follow-up
Database	✓ has <code>install_upstreams</code>
Discovery	✗ MISSING — owed in follow-up
HTTP3	✗ MISSING — owed in follow-up
Mdns	✗ MISSING — owed in follow-up
Middleware	✗ MISSING — owed in follow-up
Observability	✓ has <code>install_upstreams</code>
RateLimiter	✗ MISSING — owed in follow-up
Recovery	✗ MISSING — owed in follow-up
Security	✓ has <code>install_upstreams</code>
Tracker-SDK	✗ MISSING — owed in follow-up

Resolution path: each submodule gains its own `install_upstreams.sh` in a per-submodule commit. Per-submodule cycle: write script → commit to submodule → push to GitHub + GitLab (per §6.W) → bump pin in parent Lava repo. The parent commit can batch the 10 pin bumps once each submodule has its own `install_upstreams`.

Cross-references

- HelixConstitution Constitution.md §11.4.35 + §11.4.36 (the mandates)
- docs/plans/2026-05-15-constitution-compliance.md Phase 8
- docs/helix-constitution-gates.md CM-CANONICAL-ROOT-CLARITY + CM-INSTALL-UPSTREAMS-RAN rows

- `scripts/verify-all-constitution-rules.sh` (wraps this scanner per §11.4.32)

2026-05-15 update — HelixQA waiver

Phase 4 of the constitution-compliance plan adopted `submodules/helixqa` (HelixDevelopment-owned QA orchestration framework) at upstream HEAD `403603db`. HelixQA's `CLAUDE.md` / `AGENTS.md` / `CONSTITUTION.md` follow the canonical-root `## INHERITED FROM Helix Constitution` pointer pattern (HelixDevelopment-authored) rather than Lava's heading-anchored `§6.R` / `§6.S` / `§6.X` / `§6.AD` pointer-block format. HelixQA also lacks `helix-deps.yaml` + `install_upstreams.sh` wrapper script.

Resolution: `HELIX_DEV_OWNED=("HelixQA")` waiver list + `is_helix_dev_owned()` helper skip HelixQA in every per-Submodule loop in this scanner. Waiver entries cite Phase 4-debt: PR to HelixDevelopment/HelixQA upstream owed to add the missing files. Once upstream merges + Lava's pin advances to include them, HelixQA can be removed from the waiver list.

2026-05-16 update — HelixQA waiver RESOLVED

The Phase 4-debt PR to HelixDevelopment/HelixQA upstream landed as commit `b13ba7c` (`feat(gov): add helix-deps.yaml + install_upstreams.sh wrapper`), adding both missing files at HelixQA's repo root. Lava's pin advanced to `b13ba7c` in the same parent commit that removed HelixQA from `INSTALL_UPSTREAMS_WAIVERS`. The waiver list is now empty — the per-Submodule loop in this scanner treats HelixQA on equal terms with the other 16 owned submodules; all 17 must ship a recognizable `install_upstreams.sh` wrapper invoking `Upstreams/*.sh` (or `upstreams/*.sh`) recipes. The `INSTALL_UPSTREAMS_WAIVERS=()` empty array is intentionally retained as a forensic anchor. The §11.4.35 canonical-root inheritance pointer audit for HelixQA's `CLAUDE.md`/`AGENTS.md`/`CONSTITUTION.md` remains unaffected by this closure — HelixQA's HelixDevelopment-authored `## INHERITED FROM Helix Constitution` pointer pattern continues to satisfy the constitution-submodule's own inheritance discipline, and the §11.4.35 (a) check passes for HelixQA without needing waivers (its `CLAUDE.md` opens with the inheritance pointer, just sourced from HelixDevelopment's pattern rather than Lava's `§6.AD` block — both are valid).